Delete a Sentence Only When Putting It Back Leaves the Answer Still

Working Title

Delete a Sentence Only When Putting It Back Leaves the Answer Still.

Problem Statement

SketchThinker-R1's style reward scores a numbered list as sufficient and a long trace as not, without consulting the probability of the answer. A deleted sentence that was the only justification can leave the answer unchanged on the page while the model was not using it.

Proposed Solution

No new reward is trained until a pilot succeeds. On paired long traces and sketches, each omitted sentence is inserted back at a fixed position, and the absolute change in the answer's log probability is recorded. An unrelated sentence of similar length, inserted at the same position, is the control for a change caused by position alone. The score must predict sketch errors better than length. If it does not, probability movement is not used as a deletion rule. Only if the pilot succeeds does a second stage, with three seeds, allow a compressor to drop a sentence when restoration leaves the log probability within a pre-set threshold. The image is not edited, and steps are not scored by a separate process reward.

Key Research Questions

- Does the change on restoration predict errors better than length, above the unrelated-sentence control?
- Is the change larger for omitted visual premises than for rewritten arithmetic?
- When the filter is added, does matched-length accuracy rise only in runs whose pilot succeeded?

Expected Outcome

The study will either deliver a deletion test based on the model's own answer probability, or show that this test should not be used.

Comparison

Insertion	What it changes	What it is for
Omitted sentence	One deleted sentence, fixed site	Load-bearing check
Unrelated sentence	Similar length, same site	Control for position
Length alone	No insertion	Baseline predictor